

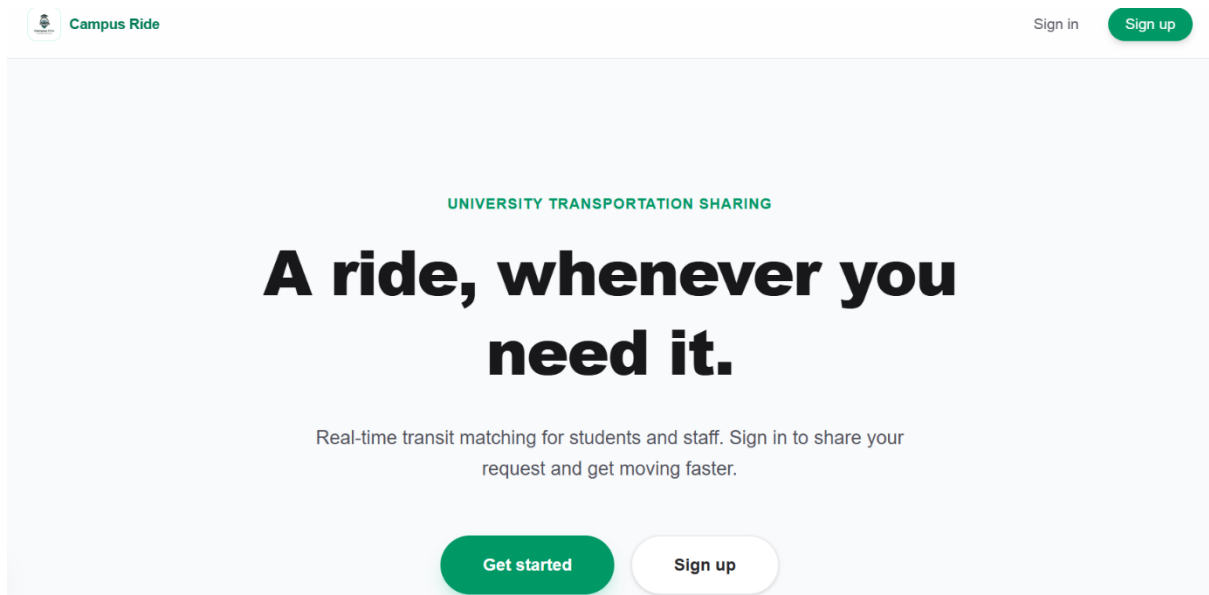
1.1-fastapi-swagger.png

Campus Ride API 0.2.0 OAS 3.1

/openapi.json

Students		^
POST	/api/v1/students/ Create Student	▼
GET	/api/v1/students/ Read Students	▼
GET	/api/v1/students/{matric_id} Read Student	▼
Drivers		^
POST	/api/v1/drivers/ Create Driver	▼
GET	/api/v1/drivers/ Read Drivers	▼
GET	/api/v1/drivers/{driver_id} Read Driver	^
Parameters		Try it out

1.2-frontend-login.png



1.3-both-terminals.png

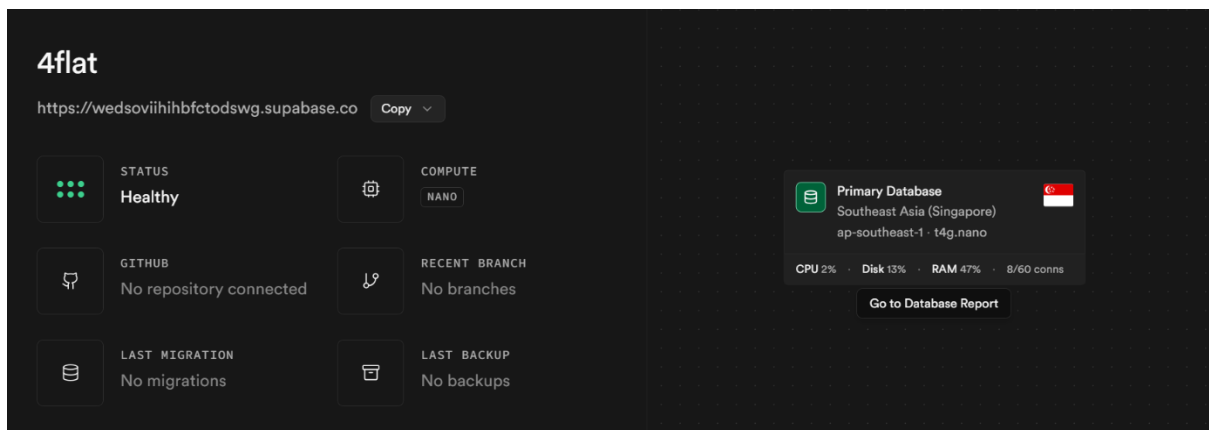
```
INFO:      Stopping reloader process [20000]
(.venv) PS C:\Users\fauzi\dist_sys_app\backend> python server.py
🚀 Starting Campus Ride Backend Server on http://127.0.0.1:8000
INFO:      Will watch for changes in these directories: ['C:\\Users\\fauzi\\dist_sys_app\\backen
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:      Started reloader process [16328] using StatReload
INFO:      Started server process [24832]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
2026-06-13 20:49:12,593 | INFO | [L7_APPLICATION] | API_REQUEST_IN | Method: GET | Path: /docs
2026-06-13 20:49:12,595 | INFO | [L6_PRESENTATION] | DE-ENCAPSULATION | Received Data PDU (Size
2026-06-13 20:49:12,599 | INFO | [L4_TRANSPORT] | SEGMENT_COMPLETE | Total RTT (Processing): 6.
INFO:      127.0.0.1:58799 - "GET /docs HTTP/1.1" 200 OK
2026-06-13 20:49:13,206 | INFO | [L7_APPLICATION] | API_REQUEST_IN | Method: GET | Path: /opena
2026-06-13 20:49:13,209 | INFO | [L6_PRESENTATION] | DE-ENCAPSULATION | Received Data PDU (Size
2026-06-13 20:49:13,376 | INFO | [L4_TRANSPORT] | SEGMENT_COMPLETE | Total RTT (Processing): 16:
```

```
(.venv) PS C:\Users\fauzi\dist_sys_app\frontend> npm run dev

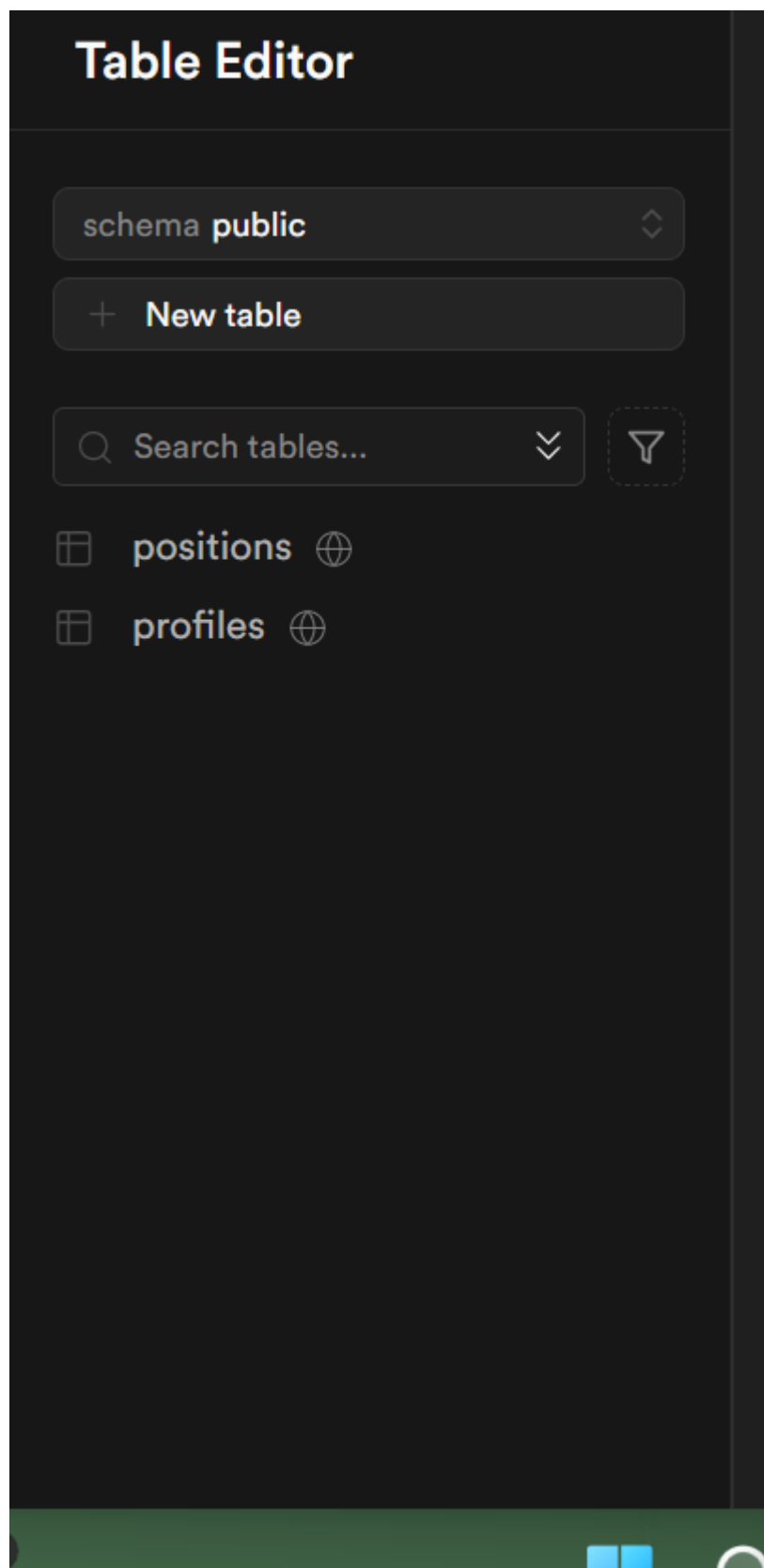
> frontend@0.1.0 dev
> next dev

⚠ Port 3000 is in use by process 13352, using available port 3001 instead.
▲ Next.js 16.2.2 (Turbopack)
- Local:      http://localhost:3001
- Network:    http://172.27.48.1:3001
- Environments: .env.local
✓ Ready in 982ms
```

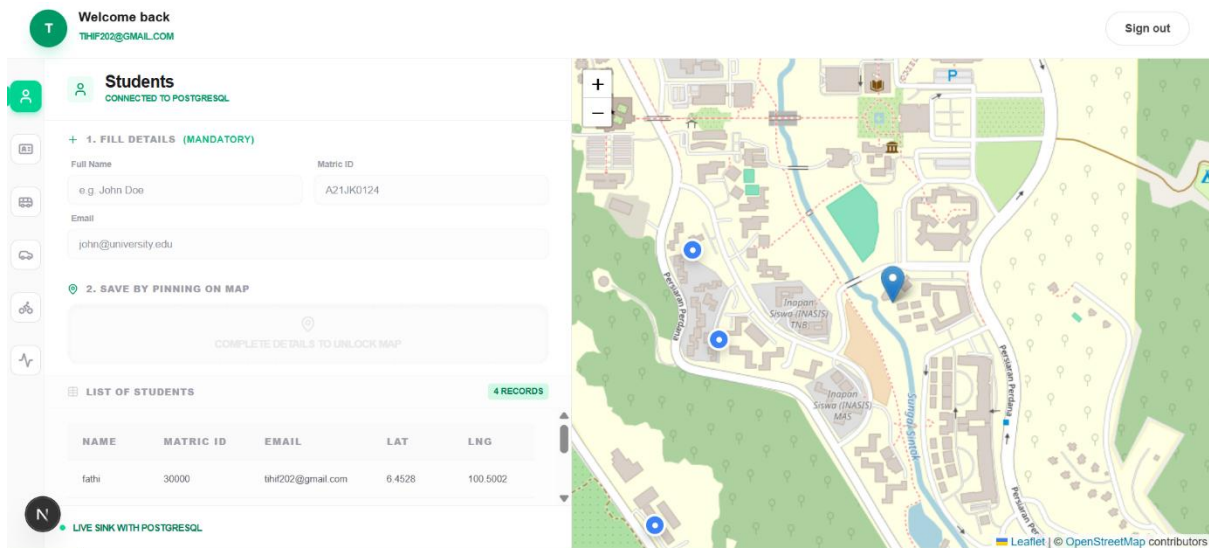
2.1-supabase-dashboard.png



2.2-supabase-tables.png



2.3-successful-login.png



3.1-docker-ps.png

```
PS C:\Users\fauzi\dist_sys_app> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
858315f9ece2   dist_sys_app-frontend              "docker-entrypoint.s..." 45 seconds ago Up 33 seconds 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp campus-frontend
218e2dfc6b5e   postgres:15-alpine                "docker-entrypoint.s..." 46 seconds ago Up 44 seconds (healthy) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp campus-postgres
```

3.2-docker-logs.png

```
PS C:\Users\fauzi\dist_sys_app> docker-compose logs --tail=30
campus-frontend > frontend@0.1.0 start
campus-frontend > next start
campus-frontend ▲ Next.js 16.2.2
campus-frontend - Local: http://localhost:3000
campus-frontend - Network: http://172.19.0.4:3000
campus-frontend ✓ Ready in 307ms
campus-backend module = importlib.import_module(module_str)
campus-backend File "/usr/local/lib/python3.11/importlib/_init_.py", line 126, in import_module
campus-backend return _bootstrap.gcd_import(name[level:], package, level)
campus-backend File "<frozen importlib._bootstrap>", line 1204, in _gcd_import
campus-backend File "<frozen importlib._bootstrap>", line 1176, in _find_and_load
campus-backend File "<frozen importlib._bootstrap>", line 1147, in _find_and_load_unlocked
campus-backend File "<frozen importlib._bootstrap>", line 690, in _load_unlocked
campus-backend File "<frozen importlib._bootstrap_external>", line 940, in exec_module
campus-backend File "<frozen importlib._bootstrap>", line 241, in _call_with_frames_removed
campus-backend File "/app/app/main.py", line 4, in <module>
campus-backend from src.infrastructure.database import engine, Base
campus-backend File "/app/src/infrastructure/database.py", line 19, in <module>
campus-backend engine = create_async_engine(
campus-backend File "/usr/local/lib/python3.11/site-packages/sqlalchemy/ext/asyncio/engine.py", line 120, in create_async_engine
campus-backend sync_engine = _create_engine(url, **kw)
campus-backend waiting for server to start...2026-06-12 09:06:42.984 UTC [41] LOG: starting PostgreSQL 15.18 on x86_64-pc-linux
-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
```

3.3-app-running.png

```
PS C:\Users\fauzi\dist_sys_app> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
campus-backend-8b89878f9-9nmz6     1/1     Running   3 (13m ago) 14m
campus-backend-8b89878f9-dnrm4     1/1     Running   1 (13m ago) 14m
campus-frontend-855dcc7b85-cn2q2   1/1     Running   0           14m
campus-frontend-855dcc7b85-wjnsv   1/1     Running   0           14m
campus-postgres-748f4c497b-rldf2   1/1     Running   0           14m
```

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4.1-kubectl-pods.png

```
PS C:\Users\fauzi\dist_sys_app> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
campus-backend-8b89878f9-9nmz6      1/1     Running   3 (13m ago) 14m
campus-backend-8b89878f9-dnrm4      1/1     Running   1 (13m ago) 14m
campus-frontend-855dcc7b85-cn2q2    1/1     Running   0           14m
campus-frontend-855dcc7b85-wjnsv    1/1     Running   0           14m
campus-postgres-748f4c497b-rldf2    1/1     Running   0           14m
```

4.2-kubectl-services.png

```
PS C:\Users\fauzi\dist_sys_app> kubectl get services
NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
campus-backend-service             ClusterIP    10.96.186.212 <none>        8000/TCP   21m
campus-frontend-service            ClusterIP    10.96.32.187  <none>        3000/TCP   21m
campus-postgres-service            ClusterIP    10.96.131.253 <none>        5432/TCP   21m
kubernetes                         ClusterIP    10.96.0.1     <none>        443/TCP    22m
```

4.3-kubectl-hpa.png

```
PS C:\Users\fauzi\dist_sys_app> kubectl get hpa
NAME                                REFERENCE          TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
campus-backend-hpa                 Deployment/campus-backend  cpu: <unknown>/70%  2         10        2           21m
PS C:\Users\fauzi\dist_sys_app> |
```

4.4-kubectl-describe.png

```

PS C:\Users\fauzi\dist_sys_app> kubectl describe deployment campus-backend
Name: campus-backend
Namespace: default
CreationTimestamp: Sat, 13 Jun 2026 12:51:47 +0800
Labels: <none>
Annotations: deployment.kubernetes.io/revision: 1
Selector: app=campus-backend
Replicas: 2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels: app=campus-backend
  Containers:
    backend:
      Image: dist_sys_app-backend:latest
      Port: 8000/TCP
      Host Port: 0/TCP
      Liveness: http-get http://:8000/health delay=30s timeout=1s period=10s #success=1 #failure=3
      Readiness: http-get http://:8000/health delay=15s timeout=1s period=5s #success=1 #failure=3
  Environment:
    DATABASE_URL: <set to the key 'database_url' in secret 'campus-ride-secret'> Optional: false
  Mounts: <none>
  Volumes: <none>
  Node-Selectors: <none>
  Tolerations: <none>
  Node-Selectors: <none>
  Tolerations: <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available       True    MinimumReplicasAvailable
  Progressing     True    NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet: campus-backend-8b89878f9 (2/2 replicas created)
Events:
  Type           Reason              Age             From              Message
  ----           -
  Normal         ScalingReplicaSet   21m            deployment-controller Scaled up replica set campus-backend-8b89878f9 from 0 to 2
PS C:\Users\fauzi\dist_sys_app>

```

4.5-port-forward-app.png

```

PS C:\Users\fauzi\dist_sys_app> kubectl port-forward svc/campus-frontend-service 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000

```

Welcome back
thif202@gmail.com

Sign out

Students
CONNECTED TO POSTGRES

+ 1. FILL DETAILS (MANDATORY)

Full Name: e.g. John Doe
Matric ID: A21JK0124
Email: john@university.edu

2. SAVE BY PINNING ON MAP

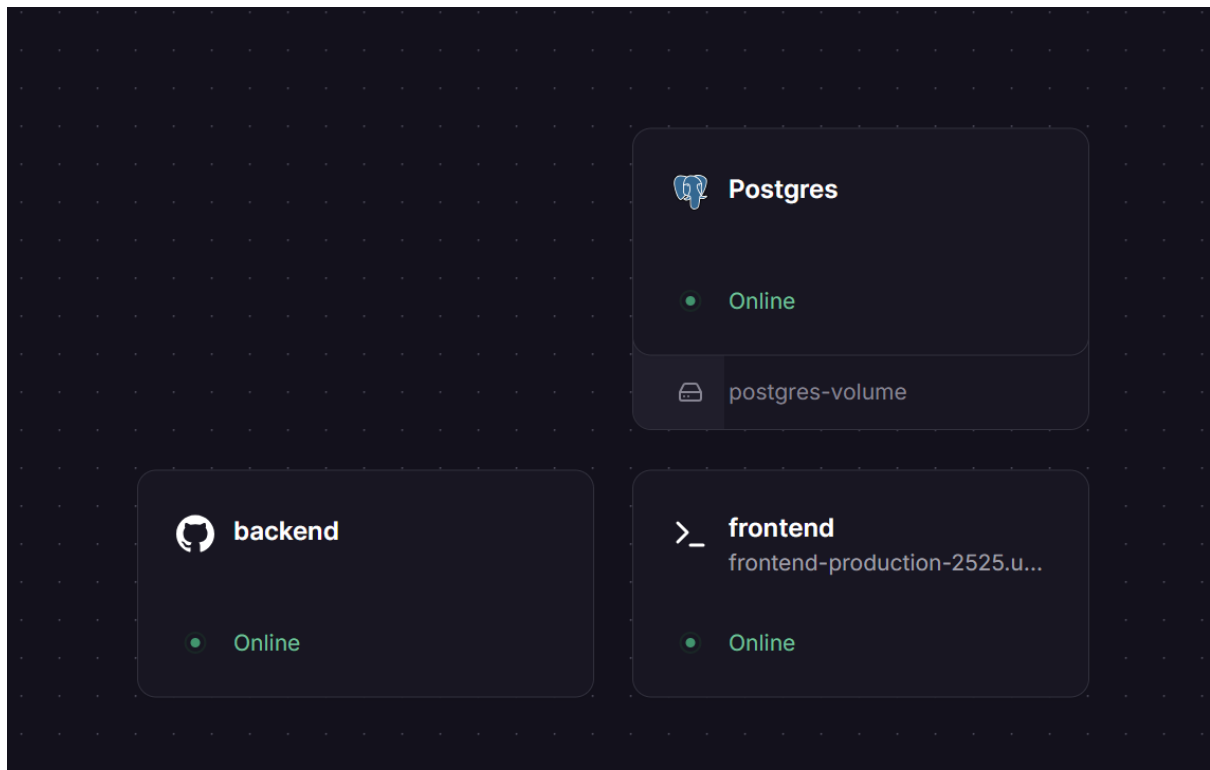
COMPLETE DETAILS TO UNLOCK MAP

LIST OF STUDENTS 4 RECORDS

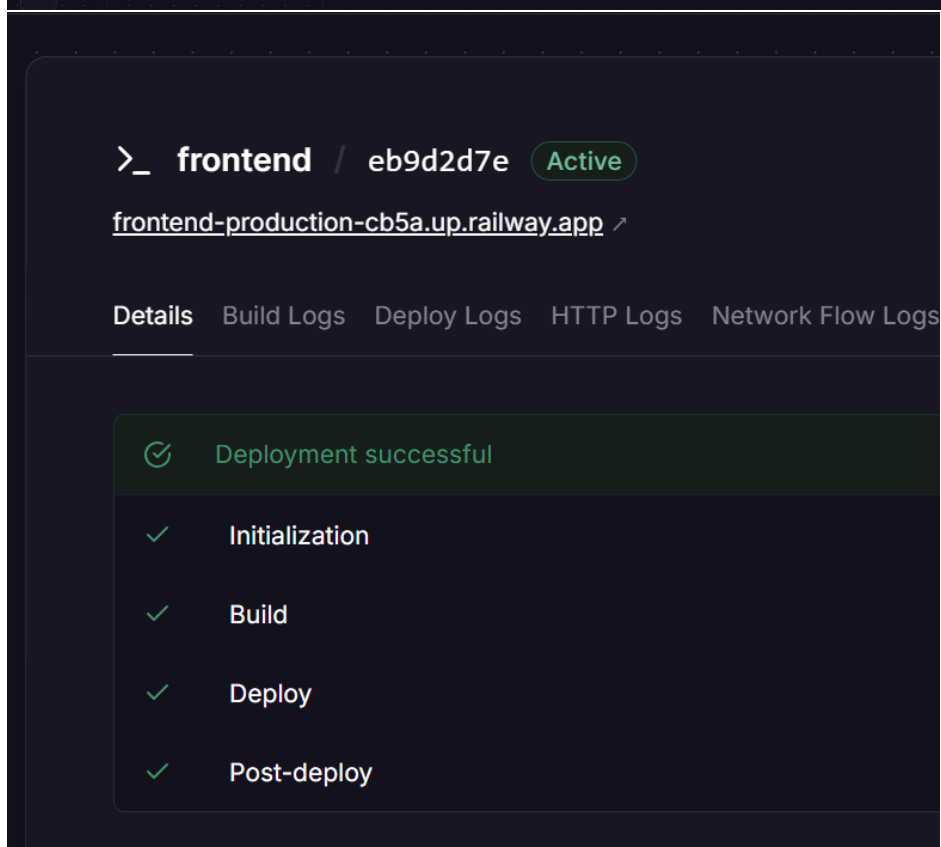
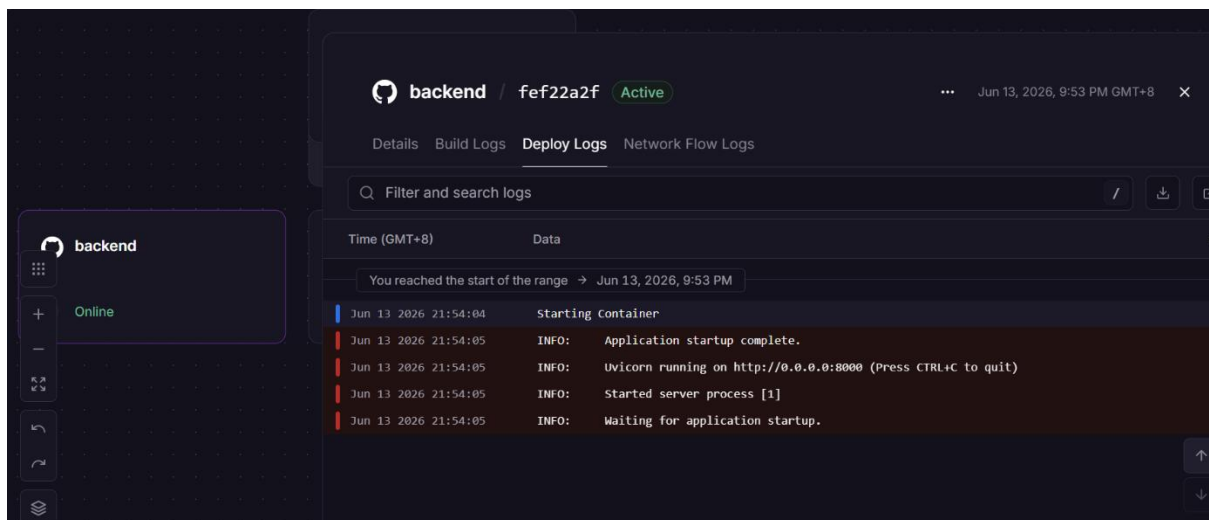
NAME	MATRIC ID	EMAIL	LAT	LNG
fathi	30000	thif202@gmail.com	6.4528	100.5002

LIVE LINK WITH POSTGRES

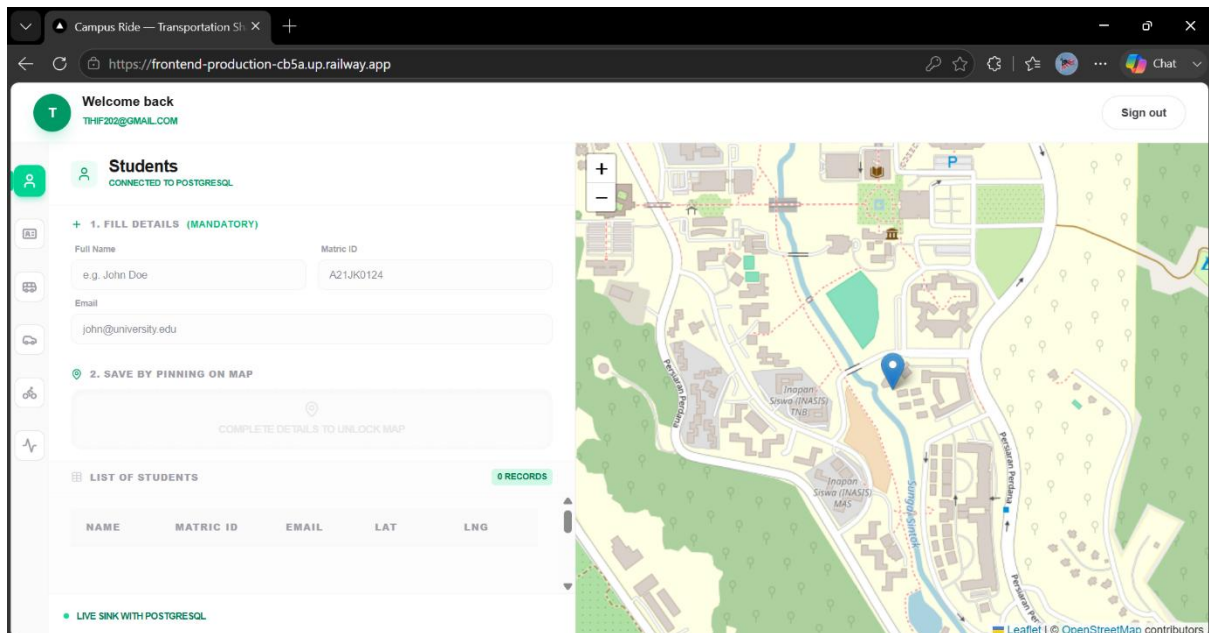
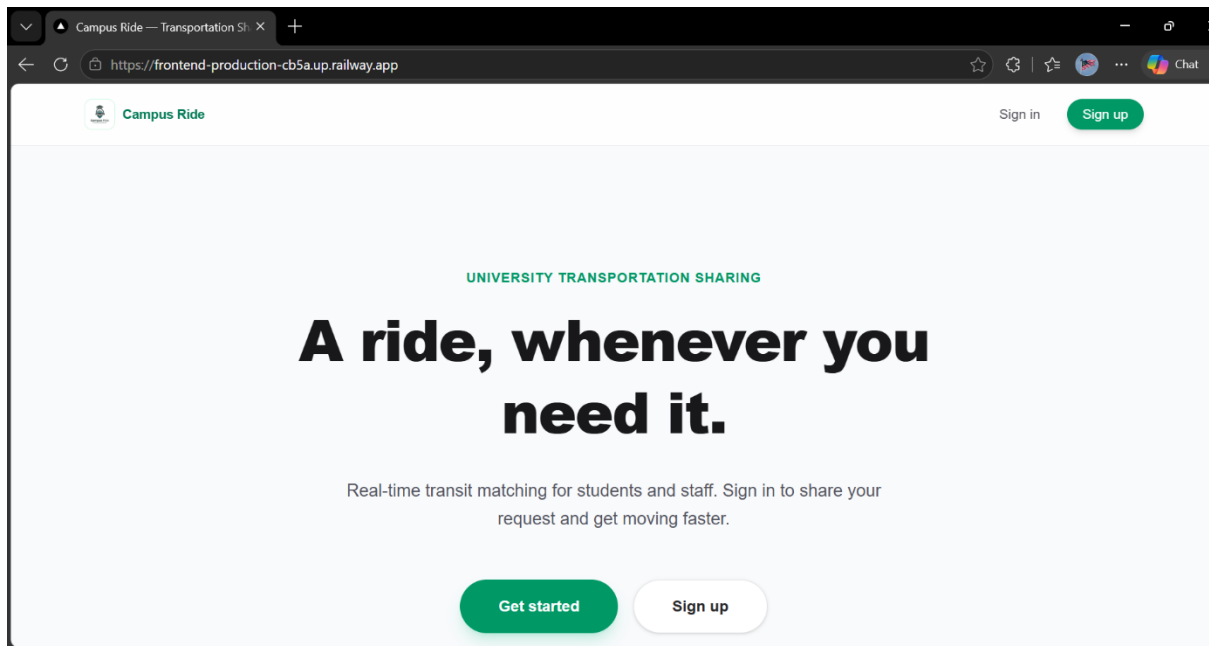
5.1-railway-dashboard.png



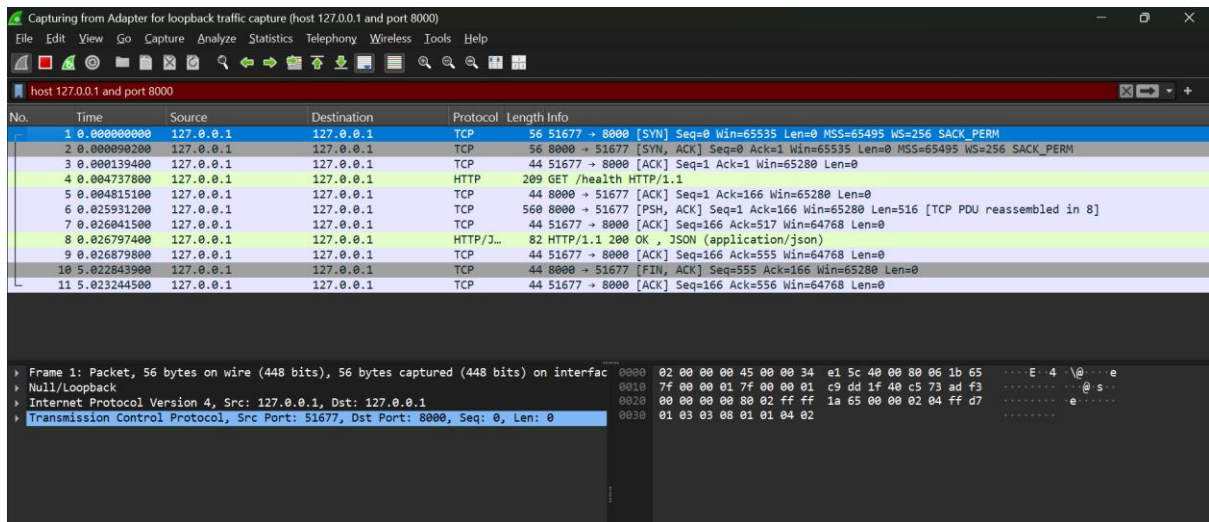
5.2-railway-logs.png



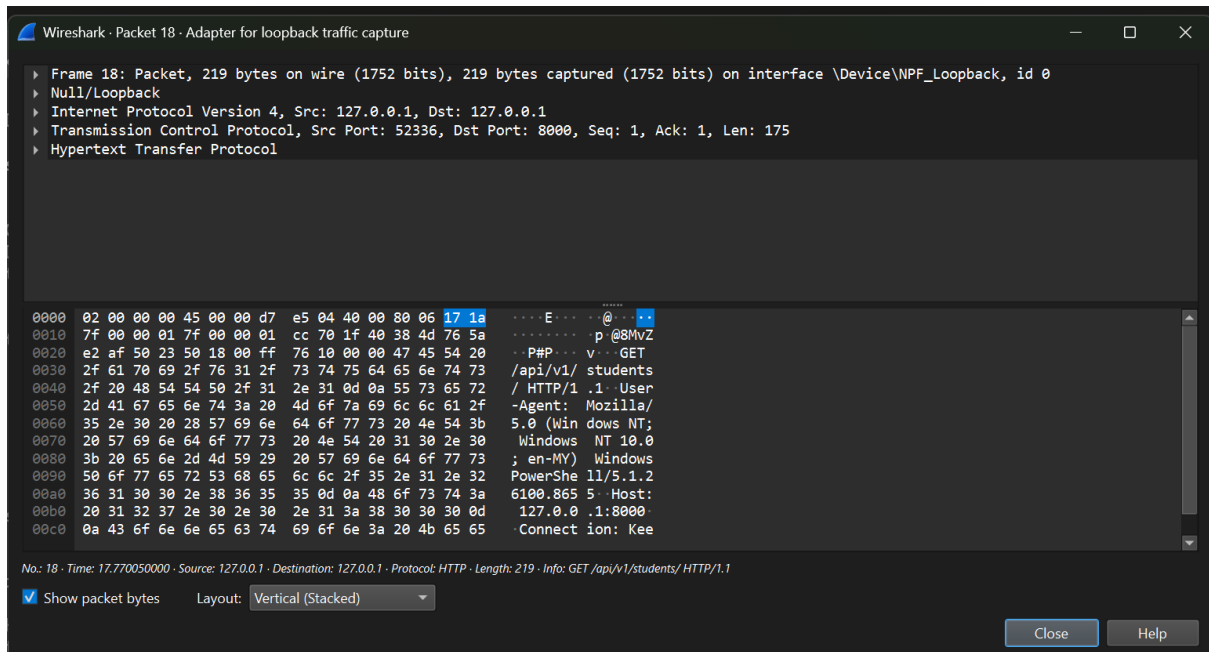
5.3-live-app.png



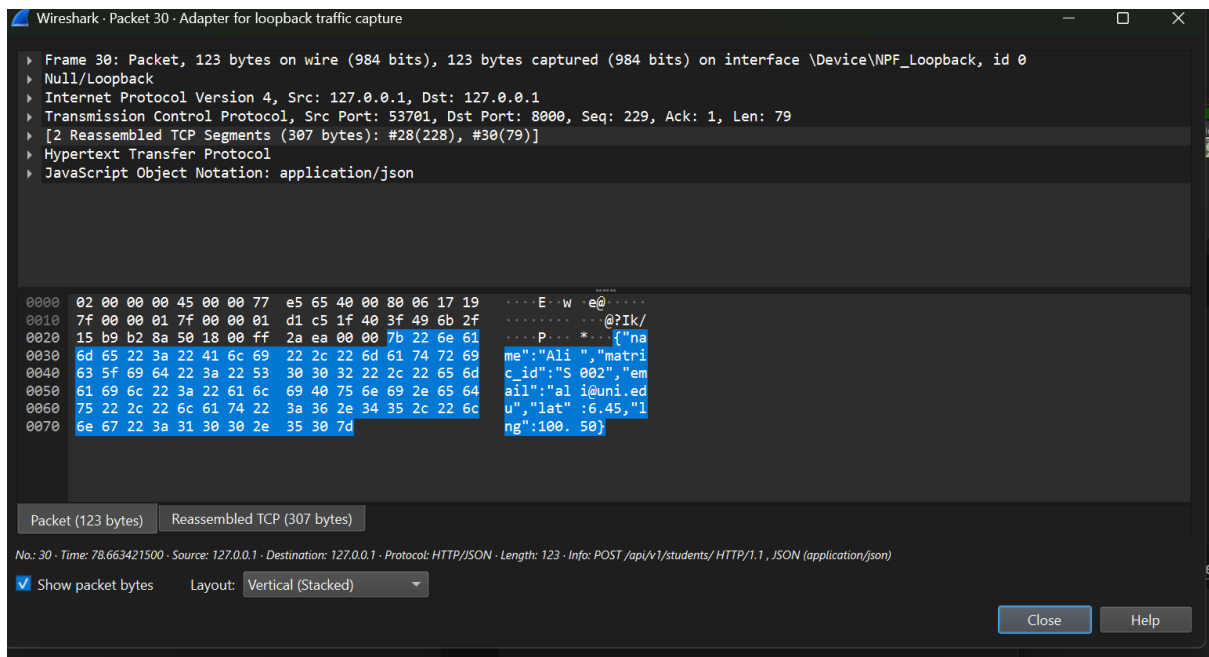
6.1-tcp-handshake.png



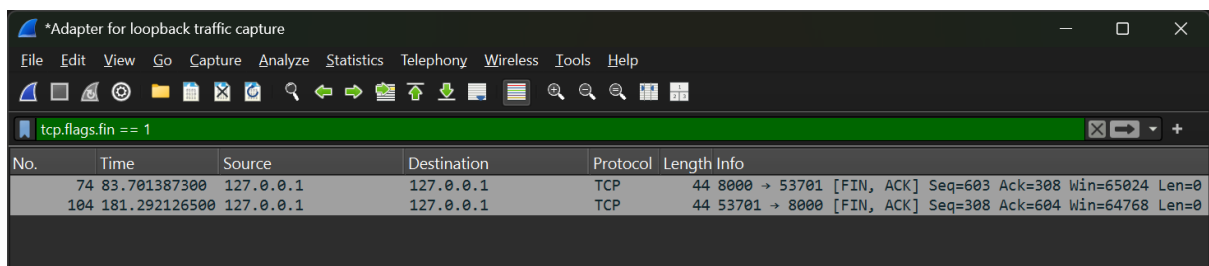
6.2-http-get-layers.png



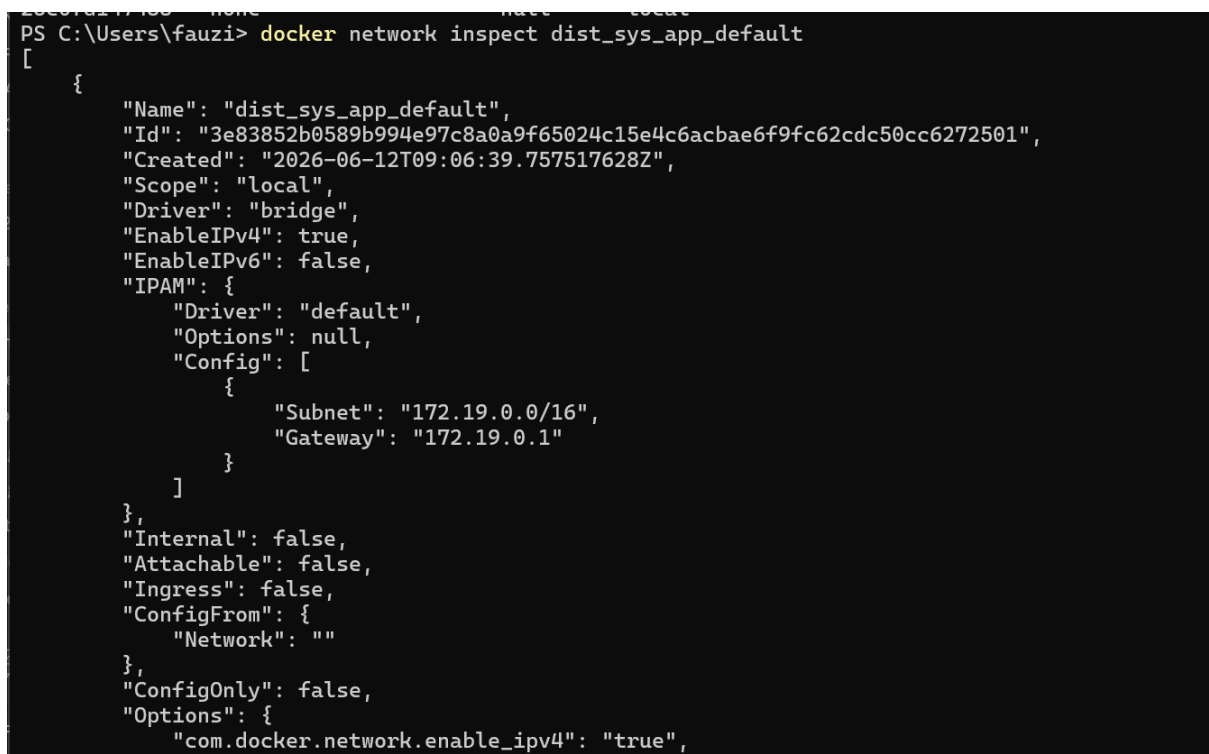
6.3-http-post-json.png



6.4-tcp-fin.png



6.5-docker-network.png



```

PS C:\Users\fauzi> docker network inspect dist_sys_app_default
[
  {
    "Name": "dist_sys_app_default",
    "Id": "3e83852b0589b994e97c8a0a9f65024c15e4c6acbae6f9fc62cdc50cc6272501",
    "Created": "2026-06-12T09:06:39.757517628Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",

```

6.6-tls-handshake.png

No.	Time	Source	Destination	Protocol	Length	Info
82	10.036813800	10.19.88.97	34.227.167.194	TLSv1.3	328	Client Hello (SNI=api.docker.com)
85	10.351005700	34.227.167.194	10.19.88.97	TLSv1.3	1304	Server Hello, Change Cipher Spec, Application Data
210	17.714147100	10.19.88.97	142.250.100.94	QUIC	1292	Initial, DCID=b2b9ff31f6e1cb07, PKN: 1, CRYPTO, PING, CRYPTO, PING, CRYPTO, PADDING, CRYPTO,
211	17.714575300	10.19.88.97	142.250.100.94	QUIC	1292	Initial, DCID=b2b9ff31f6e1cb07, PKN: 2, PING, PING, CRYPTO, CRYPTO, PING, CRYPTO, PING, PING,
217	18.025759600	142.250.100.94	10.19.88.97	QUIC	1292	Initial, SCID=f2b9ff31f6e1cb07, PKN: 3, CRYPTO, PADDING
244	20.247031400	10.19.88.97	23.55.39.11	QUIC	1292	Initial, DCID=5f9c47142d5e9466, PKN: 2, CRYPTO, PING, CRYPTO, PING, PING, PADDING, PING, PADD
251	20.375534100	23.55.39.11	10.19.88.97	QUIC	1292	Initial, SCID=077cedc96340d31, PKN: 3, CRYPTO, PADDING
257	20.387668300	10.19.88.97	160.79.104.10	QUIC	1292	Initial, DCID=a225b58869666cfa, PKN: 2, CRYPTO, PING, CRYPTO, PING, PING, CRYPTO, CRYPTO, PING, PADD
268	20.416911300	160.79.104.10	10.19.88.97	QUIC	1242	Initial, SCID=0150f5cef9b1dca950cdce28f9b9192f75a1b5, PKN: 1, ACK, CRYPTO
269	20.417674400	160.79.104.10	10.19.88.97	QUIC	1242	Handshake, SCID=0150f5cef9b1dca950cdce28f9b9192f75a1b5
384	24.680346800	10.19.88.97	20.42.73.30	TLSv1.3	677	Client Hello (SNI=functional.events.data.microsoft.com)
386	24.682117700	10.19.88.97	20.42.73.30	TLSv1.3	741	Client Hello (SNI=functional.events.data.microsoft.com)
388	24.985564800	20.42.73.30	10.19.88.97	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
390	24.985564800	20.42.73.30	10.19.88.97	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
391	24.988076900	10.19.88.97	20.42.73.30	TLSv1.3	773	Change Cipher Spec, Client Hello (SNI=functional.events.data.microsoft.com)

Frame 82: Packet, 328 bytes on wire (2624 bits), 328 bytes captured (2624 bits) on int	0000	50 49 21 a7 10 5f 10 68 38 fd 28 42 08 00 45 00	PII...h 8 (B-E
Ethernet II, Src: AzureWaveTec_fd:28:42 (10:68:38:fd:28:42), Dst: Cisco_a7:10:5f (50:4	0010	01 3a 3b e4 40 00 80 06 90 c0 0a 13 58 61 22 e3	;;@...Xa"
Internet Protocol Version 4, Src: 10.19.88.97, Dst: 34.227.167.194	0020	a7 c2 cc 6a 01 bb ac 4d 08 1b 87 47 80 ea 50 18	...j...M...G P
Transmission Control Protocol, Src Port: 52330, Dst Port: 443, Seq: 1251, Ack: 1, Len:	0030	00 ff f0 fb 00 00 06 70 b1 ce 9c 96 94 b9 6e 9d	...p...n
[2 Reassembled TCP Segments (1524 bytes): #81(1250), #82(274)]	0040	48 92 a9 12 59 0b b4 bf af 15 7f 3c 7b b0 99 95	H...Y...<{...
Transport Layer Security	0050	02 8c f8 b2 bd 2a 6e 5c f1 b1 21 7a cc a3 a1 c5	...n\...iz...
	0060	7c 9c 74 48 60 59 ca c1 2e bc 41 12 8f 6c 2e ee	tH'Y...A-1..
	0070	15 47 16 23 8c 9a 9b 47 c3 99 5b 56 f7 b7 57 52	G #...G [V-WR
	0080	c0 5b e7 a3 40 11 a6 f2 8c 5c a9 ea 0c 28 08 cb	[...@...V...<...
	0090	77 b7 98 0b 12 50 8d 11 b6 48 71 2d c9 a5 00 20	w...P...Hq...
	00a0	42 75 8d 4c 0a 6c c6 5e c5 44 ba d2 7c c5 e9 b1	Bu L 1...D ...
	00b0	cc 12 2b 41 8a 04 52 5e 74 c2 11 04 89 dd e6 53	+A R" t...S
	00c0	d0 63 1d e7 6b cd d3 79 28 2d a6 2b a9 30 a0 ad	-c-k-y (-+0-

7.1-artillery-results.png

```
Test run id: t59qe_kya8zjc7g97m3e9f7x6a8dpgrany_jcmn
Phase started: Phase 1 - Low load (index: 0, duration: 60s) 11:31:38(+0800)
```

```
-----
Metrics for period to: 11:31:40(+0800) (width: 0.126s)
-----
```

```
http.request_rate: ..... 2/sec
http.requests: ..... 2
vusers.created: ..... 3
vusers.created_by_name.Health check and students: ..... 3
```

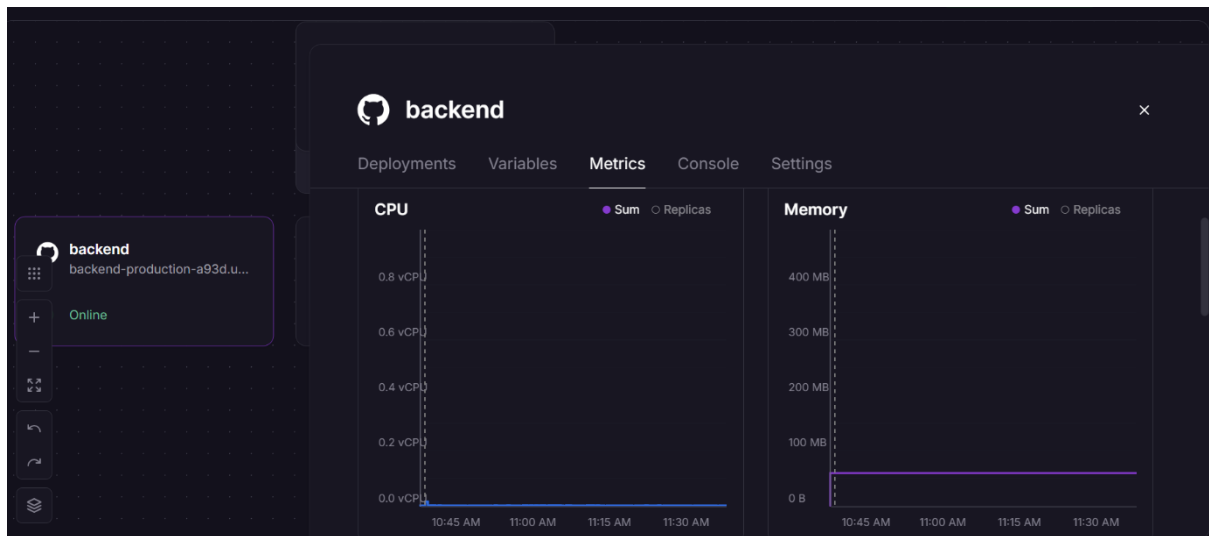
```
-----
Metrics for period to: 11:31:50(+0800) (width: 9.983s)
-----
```

```
http.codes.502: ..... 199
http.downloaded_bytes: ..... 21691
http.request_rate: ..... 20/sec
http.requests: ..... 201
http.response_time:
  min: ..... 255
  max: ..... 941
  mean: ..... 284.7
  median: ..... 267.8
  p95: ..... 295.9
  p99: ..... 757.6
```

```
http.downloaded_bytes: ..... 1294188
http.request_rate: ..... 37/sec
http.requests: ..... 13200
http.response_time:
  min: ..... 252
  max: ..... 1283
  mean: ..... 277.3
  median: ..... 267.8
  p95: ..... 290.1
  p99: ..... 507.8
http.response_time.5xx:
  min: ..... 252
  max: ..... 1283
  mean: ..... 277.3
  median: ..... 267.8
  p95: ..... 290.1
  p99: ..... 507.8
http.responses: ..... 13200
vusers.completed: ..... 6600
vusers.created: ..... 6600
vusers.created_by_name.Health check and students: ..... 6600
vusers.failed: ..... 0
vusers.session_length:
  min: ..... 574.6
  max: ..... 3655.7
  mean: ..... 642.8
  median: ..... 608
  p95: ..... 804.5
  p99: ..... 1408.4
```

```
PS C:\Users\fauzi\dist_sys_app> |
```

7.2-railway-cpu-spike.png



7.3-kubectl-hpa.png

```
PS C:\Users\fauzi\dist_sys_app> kubectl get hpa
NAME                REFERENCE                TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
campus-backend-hpa  Deployment/campus-backend  cpu: <unknown>/70%    2         10        2          22h
PS C:\Users\fauzi\dist_sys_app> |
```